Symbiosis Institute of Technology · Department of AIML

GuardianEye: Late Fusion of Image and Audio Poaching Indicators

Big Data Analytics (BDA) · Course code 0701260704 · Academic Year 2026–2027

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Proof-of-concept. Indicators, not confirmed poaching.

Problem and Motivation

- Wildlife patrol evidence arrives as camera-trap RGB, aerial thermal frames, and short audio clips.
- No labelled poacher class in the collections named by configuration.
- No public time-aligned field incidents pairing image with audio of the same moment.
- GuardianEye detects indicators (human / vehicle / gunshot-like sound), not confirmed poaching.
- One detection is not proof of poaching; late fusion informs a human review workflow.
- Video is not a third modality: frames are sampled into the image branch.

Objectives (Completed Work)

- Implement a two-branch indicator pipeline: YOLOv8 image domains + log-mel audio CNN.
- Keep image metrics separate for aerial_thermal and camera_trap_rgb.
- Stratify audio by site (Belize ↔️ Vietnam) with in-domain controls.
- Retain empty images and background audio; address class imbalance (oversampling / weighted loss).
- Define late-fusion rules and evaluate fusion pairs under a labelled synthetic proxy protocol.
- Serve the pipeline (FastAPI + Gradio dashboard) with SQLite audit logging.
- Aggregate prepared manifests and the audit log with a local PySpark Big Data Analytics job.

System Architecture

Image branch

- Still / sampled video frames
- Domains: aerial_thermal, camera_trap_rgb
- YOLOv8 per domain
- Classes: human, vehicle, animal, empty

Audio branch

- WAV windows or spectrogram image
- Fixed windows → log-mel
- Small CNN classifier
- Classes: gunshot, background

Decision & serving

- Late-fusion rules (HIGH / MEDIUM / LOW)
- FastAPI API + Gradio dashboard
- SQLite event / audit log
- Alerts and non-alerts both recorded

HIGH requires human/vehicle and gunshot-like score within the time window. Single-modality uploads cannot reach HIGH.

Data Pipeline and BDA Framing

- Prepared manifests under data/processed/ (JSONL); configuration in configs/*.yaml; seed 27.
- Dataset download URLs in configs/data.yaml remain TODO — not invented.
- Empty images retained: 12 of 22; background audio retained: 20 of 29.
- Class imbalance (max/min): aerial_thermal 6.0, camera_trap_rgb 6.0; audio Belize 2.5, Vietnam 2.0.
- Handling: rare visual classes oversampled; audio uses class-weighted loss.
- BDA artefact aggregates indicator metadata counts — not detector mAP / F1 scores.

Corpus	Records	Key retention	Imbalance note
Images	22	12 empty kept	ratio 6.0 per domain
Audio windows	29	20 background kept	Belize 2.5 / Vietnam 2.0

Audio Site Stratification

Belize ↔️ Vietnam cross-site transfers and in-domain controls

Experiment	Train	Val	Test
Belize → Vietnam	11	3	14
Vietnam → Belize	11	3	14
In-domain Belize	8	3	3
In-domain Vietnam	8	3	3

- Site counts: Belize 14 windows (gunshot 4, background 10); Vietnam 15 (gunshot 5, background 10).
- Split-assignment rows across four experiments: 84 (files: 4).
- Cross-site experiments expose domain shift; they are not averaged into one headline score.

Synthetic Proxy Fusion Protocol

Constructed pairs — not time-aligned field incidents

- Protocol label: synthetic_proxy (16 pairs).
- By domain: aerial_thermal 8, camera_trap_rgb 8.
- Expected levels: HIGH 6, MEDIUM 6, LOW 4.
- Pairs are built from labelled scenarios for review of fusion rules.

Scenario	Pairs
animal_background	2
empty_gunshot	2
empty_impulse	2
human_background	2
human_gunshot	2
small_human_gunshot	2
vehicle_background	2
vehicle_gunshot	2

Honesty note

Constructed pairs. Not time-aligned field incidents. Fusion evaluation under this protocol is a synthetic proxy, not field validation.

Big Data Analytics Program

bda/guardianeye_bda_analysis.py

- Engine: PySpark 4.0.1 on master local[2] (Java: OpenJDK 21).
- Inputs: image manifest, audio windows, site-stratified split CSVs, synthetic proxy pairs, SQLite audit log.
- Pattern: map each record to grouping keys; reduceByKey / Spark SQL aggregations for counts and ratios.
- One-hour tumbling window over the event log (timestamp → window → alert / latency aggregates).
- Outputs: reports/bda/summary.json and reports/bda/aggregation.csv.
- Scope: metadata aggregation of prepared indicator corpora — not model scoring.

1-hour audit window (from event log)

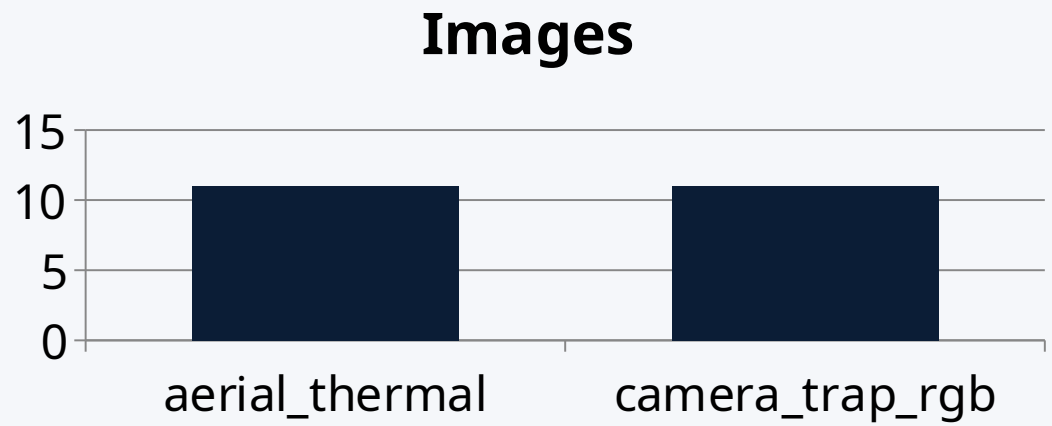
2026-10-26 10:00:00 → 2026-10-26 11:00:00 · alerted 0 · not alerted 1 · mean latency 969.8 ms

Empirical Results — Image and Audio Counts

From reports/bda/summary.json only. Not detection metrics.

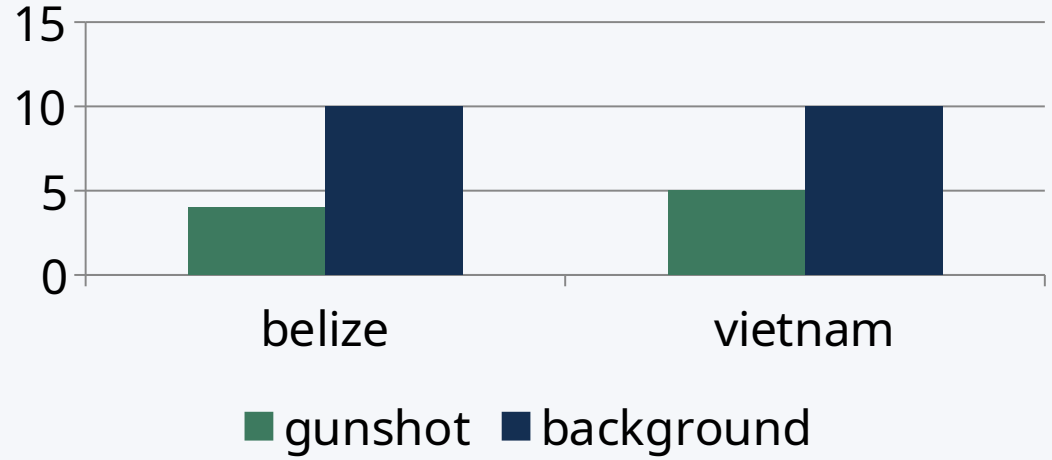
Image records by domain and class

Domain	animal	empty	human	vehicle
aerial_thermal	2	6	2	1
camera_trap_rgb	2	6	2	1



Audio windows by site and label

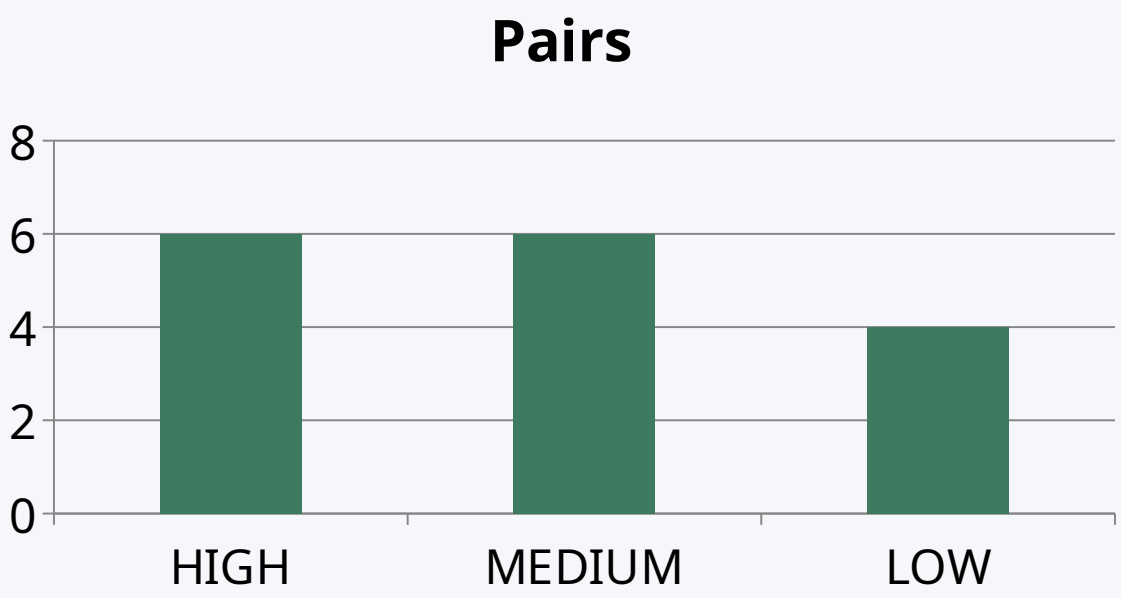
Site	gunshot	background
belize	4	10
vietnam	5	10



Empirical Results — Proxy Pairs and Audit Log

Metadata counts only. No fabricated mAP / F1 / precision / recall.

Synthetic proxy pairs by expected level



SQLite audit log (aggregated)

- Event records: 1
- Alerted: 0 · Not alerted: 1
- Risk level: LOW=1
- Domain: camera_trap_rgb=1
- Reason: unreliable_weights=1
- Proxy pairs total: 16 (protocol=synthetic_proxy)
- Seed: 27

Checkpoint status: models/ contains no trained weights. Detection metrics are not reported. summary.json note: analytical summary of prepared indicator metadata — not model metrics.

Serving and Honesty

- Service: FastAPI on port 8742; Gradio dashboard at /dashboard.
- GET /health for liveness; GET /model-info reports whether each checkpoint loaded (smoke / trained / missing).
- POST /events accepts visual and/or audio uploads; GET /events returns the audit trail.
- Every decision is logged, including non-alerts.
- Current working tree: no checkpoint under models/ — weights are unreliable / missing.
- Observed audit event reason in BDA aggregation: unreliable_weights (risk LOW; not alerted).
- Smoke checkpoints, if present, prove scripts run; they are not field performance.

Conclusion

- GuardianEye delivers a completed late-fusion indicator pipeline for wildlife evidence review.
- Image and audio branches remain domain- and site-aware; empty and background samples are retained.
- Fusion pairs are labelled as a synthetic proxy protocol — not field-validated poaching detection.
- PySpark BDA job aggregated the prepared corpus: 22 images, 29 audio windows, 84 split rows, 16 proxy pairs, 1 audit event(s).
- No detection accuracy figures are claimed without checkpoints; models/ has none in this tree.
- One detection remains an indicator, not proof of poaching.

Thank You

Questions and Discussion

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